

A CHANGELOG · EDITION 5.5

RTMcompare

Every change worth telling you about.
Newest first.

This is the receipt. Not a press release — the actual list of what changed and why. If you skipped a few releases and want to know what's different now, start here.

5.5.0 is the big one: a fresh coat of paint on the UI, a rewritten click detector, a state-of-the-art stem separator, and one big subtraction (the AI panel).

The "?" button works now.

A one-liner shipped on top of 5.5.0 to fix the click route on the new shortcut-help button.

The `?` button in the header dispatched a custom event (`rtm-toggle-shortcuts`) that nothing was listening for, so clicking it did nothing. Pressing the `?` key still toggled the shortcuts panel correctly — only the click route was broken.

ShortcutHelp.tsx now subscribes to the custom event in addition to its keyboard handler. Both routes work.

If you're on 5.5.0 and only ever press `?` instead of clicking, you don't need this update.

Design bump.

A hard look at the UI. The bits that got in your way are gone.

Reference history dropdown

Starred refs and recent picks live in a compact menu on the main page instead of a wall of names. Click, scroll, pick, go.

" + New analysis" in the header

Reset and start fresh from anywhere — no more hunting through the menu.

Search palette (DK)

Jump to anything, fast. New header button to summon it.

Keyboard-shortcut sheet (?)

All the hotkeys, one tap away. Header button.

"Level matched" pill on EQ Preview

A/B-ing the EQ tweak with and without level-matching used to take three clicks. Now it's one. Pill on the panel, not buried in a gear-icon menu.

Sticky Advanced QC

Open it once, it stays open across sessions. No more re-toggling every time you launch the app.

Quieter single-file & folder scans

Removed the Ceiling and ADM warnings that fired on every non-Atmos file — they were noise, not signal.

Click & glitch detector — drum-friendly.

The v1 click detector flagged every snare hit. Replaced it with FLOW v2 (LPC residual, Godsill & Rayner 1998).

No more drum false positives

Tuned to FLOW's strict production default (sensitivity 1.0, $K = \max(6, 12/\text{sens}) = 12$). The drums you put there on purpose stay un-flagged.

Better severity ranking

The top-20 list now sorts by severity then ratio, so the worst offenders surface first.

Cleaner deduplication

Removed the redundant 80 ms double-dedupe that was hiding adjacent real clicks.

Stem separator — BS-RoFormer.

Quiet but big. The Demucs-only separator from earlier 5.x has been replaced with BS-RoFormer 4-stem.

SDR 9.66 on MUSDB18HQ

Up from ~7.5 with Demucs. Cleaner stems = more reliable downstream analysis (Match tab, EQ Preview level-match, masking).

Auto-falls-back to Demucs

If BS-RoFormer can't load on a given machine, the pipeline drops back to Demucs. Older bundles and edge cases keep working.

AI detection.

We were going to ship UAI's 24-detector calibrated ensemble (F1 0.998 on Lambda validation). It works beautifully — but we pulled it.

The full strict engine would have added ~1.1 GB of model weights to the bundle (BS-RoFormer ckpt + CLAP audio embeddings + ONNX detectors + calibration heads) for one optional feature most engineers don't act on day-to-day. The math:

- CLAP weights — 589 MB
- UAI ONNX detectors — 46 MB
- BS-RoFormer ckpt (also used by separator) — 503 MB
- Calibration heads — < 1 MB

We're keeping BS-RoFormer (it powers stem separation everywhere in the app) and dropping the rest. The bundle gets ~840 MB lighter. May revisit AI detection as a separate opt-in download.

If you relied on the AI panel, sit on 5.4.0 for now or shout.

Smaller, cleaner bundle.

The Mac DMG drops nearly a gigabyte. The Windows zip finally ships with the separator weights it was missing in 5.4.0.

Mac DMG: -836 MB

python-bundle dropped 200 MB (xgboost / transformers / huggingface_hub / libomp out). model-cache dropped 636 MB (CLAP + UAI ONNX + calibration heads out).

Windows zip — finally complete

5.4.0's Win bundle silently shipped without the BS-RoFormer ckpt — stem separation didn't work at all on Windows. The Win CI now pre-downloads the 503 MB ckpt + yaml into model-cache before electron-builder runs.

Win CI cache key v3-uai → v5-stems-only

Forces a clean rebuild without the AI deps. Saves ~3 min on cache hits going forward.

RTMprofile production model-cache discovery

Standalone RTMprofile.app now walks up the bundle to find model-cache/ correctly — no more "model not found" on first launch from /Applications.

24 audit fixes

Mostly cosmetics, a couple of real wiring bugs caught during the pre-flight pass.

How we got here.

The shorter version of every release before this one.

5.4.0

UAI integration kickoff. BS-RoFormer 4-stem + 24-detector ensemble landed, but shipped without the strict engine deps so the panel ran in validation mode in production. Click detector v2 first cut. 24 audit fixes.

5.3.0

Vendored the UAI detector runtime (modspec, lofcz, CNN, AST). First pass at the calibrated ensemble. Mac signing/notarization automation.

5.2.x

Windows bundle reached parity with Mac (added librosa/numba/demucs/julius/openunmix). Atmos pipeline polish. Pitch deck refresh.

5.1.x

First Windows release. EBU R128 fixes. Album batch view.

5.0.x

Initial 5.x line — A/B compare, single-file QC, Atmos, the whole bundle structure.
